Empirical Proof Record: VALIDATED

2026-05-24T14:13:28.731836+00:00

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Proof ID: `5a9f1b5764d895f1fd3e8c9267c9ae30…`
Created: 2026-05-24T14:13:28.731836+00:00

1. Claim

> A learned-from-scratch ear gives a discriminating audio address on real ESC-50 sounds: held-out (source-disjoint fold 5) prec@5 >= 0.33, clearly beating raw log-mel (~0.227) and the live 5-stat encoder (~0.13 ≈ a random projection).

- **Operationalisation:** ESC-50 (50 classes × 2000 WAV); log-mel 64×128; a small from-scratch CNN ear (no pretrained weights) trained on folds 1-4; address = penultimate embedding; held-out fold-5 prec@5 (within−cross cosine NN), vs the 5-stat AudioGeoidEncoder and raw log-mel mean-pooled on the same fold.
- **Threshold:** `ear_holdout_prec_at_5 >= 0.33 precision@5`
- **H0:** H0: learned-ear prec@5 < 0.33 — a learned ear does not recover audio discrimination
- **H1:** H1: learned-ear prec@5 >= 0.33 — a faithful learned ear

2. Verdict

VALIDATED — observed `0.3405`
learned-ear held-out prec@5 0.3405 (margin 0.285) over 400 real clips, 50 classes (chance 0.020); 5-stat 0.1290 (margin 0.015 ≈ random); raw log-mel 0.2270

3. Evidence

Pillar	Statistic	Value	95% CI	Library
`audio_address_discrimination`	`ear_holdout_prec_at_5`	`0.3405`	(0.0000, 0.0000)	numpy 2.4.4

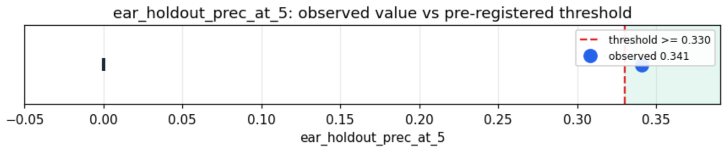

Figure 1: ear_holdout_prec_at_5 – observed value with Wilson 95% CI.

4. Pre-registration

- Registered at: `2026-05-24T14:13:28.481678+00:00`
- Config hash: `42661572305b7499123304ede00c46c3…`
- Data hash: `0c75c5335aeeb189f84648dfee816cf5…`

5. Signature

`31970a88b946c9f96eda71ddbc5645a8015bf35cd351d140670336c64fee255e`